Resident OS

Production Blueprint

A single, canonical plan for building an evidence-backed AI operations layer for apartment communities.

What it is	What ships first	What makes it credible
A decision layer beside property-management systems, not a replacement system of record.	The measured Maintenance Loop: report, safety, evidence, decision, approval, dispatch, trace.	Every consequential action carries versioned evidence, guardrail outcomes, human action, and an audit receipt.

Product thesis

Property-management software records work but rarely decides what should happen next. Resident OS acknowledges a resident issue immediately, assembles grounded evidence, proposes a safe action, requires the right human review, and keeps the proof needed to explain the result.

The product is not a more talkative bot. It is safer, explainable apartment operations.

Canonical source documents

- PRODUCT.md: user outcomes, scope, requirements, experience, and release gate.
- TECHNICAL-ARCHITECTURE.md: engineering invariants, platform, agent system, security, evaluation, and operations.
- DELIVERY-PLAN.md: one build sequence from contract freeze through production launch.

Status: planning and contract freeze. Targets in this document are targets until an evaluation run publishes reproducible results with a commit SHA.

1. Product and scope

The initial customer is a mid-market apartment operator with 1,000 to 15,000 units. Maintenance is the first loop because it produces cheap ground truth: priority correction, vendor acceptance, first-time fix, reopen rate, cost, and human override reasons.

Measured MVP

In scope	Explicitly deferred
Maintenance intake; deterministic life-safety screen; retrieval; diagnosis; policy audit; manager approval; dispatch proposal; trace; evaluation; ML operations.	Payments; rent collection; tenant screening; legal advice; health data; community feed; marketplace; autonomous high-blast-radius action.

People and success

User	Job to be done	Success condition
Resident	Report an issue and know it is handled.	Ticket number in under one second; a plain-language status and charge explanation.
Manager	Clear the queue without an unsafe or expensive mistake.	SLA risk, evidence, cost, confidence band, and approve/edit/reject controls together.
Tech or vendor	Receive and complete the right job.	Scoped job context, access and parts information, or a signed no-login link.
Owner and operations	See outcomes and trust the AI system.	Cost/SLA/governance view plus a complete audit and quality trail.

Experience rules

- The manager approval queue is the primary product and demonstration screen.
- The resident view uses the same facts but never exposes model names, prompt detail, or confidence scores.
- A status lamp is always paired with text. Colour never carries priority alone.
- Demo mode is visibly synthetic and blocks outbound transport at the adapter boundary.

2. The Maintenance Loop

A safe flow separates fast acknowledgement, deterministic safety handling, evidence assembly, decision proposal, human authority, and execution.

Stage	Behaviour	Required proof
1. Acknowledge	Persist ticket and return a number before model work.	Idempotency receipt and sub-one-second measurement.
2. Safety	Rules and a small-model second opinion combine with OR logic. P0 pages a human with no model after it.	Triggered rule/model signal and paging receipt.
3. Evidence	Collect versioned SOP, lease, SLA, unit, asset, warranty, and vendor evidence in parallel.	Citation IDs, scores, source version, effective date, and authority.
4. Propose	Typed diagnosis, priority, responsible party, and dispatch plan.	Validated schema, factual claims cited or declared unknown.
5. Audit	Deterministic policy precedence and confidence gate decide auto, review, or abstention.	Guardrail and policy verdicts, confidence rationale, escalation reason.
6. Authorise	Manager approves, edits, reassigns, or rejects.	Actor, action, time, and reject/override reason.
7. Execute	Scoped tool creates the work order or communication.	Single-use authority and external execution receipt.
8. Learn	Asynchronous judge and operations loop score the completed run.	Pinned evaluation version; labelled failures enter review, not training automatically.

Safety, authority, and transparency

- Only the orchestrator owns write tools. All other components read, analyse, or propose.
- A notification policy state machine, not a model, decides whether, when, and how to contact people.
- A meaningful decision remains explainable as a trace for an operator and as plain language for a resident.

3. Architecture

The product uses a clear separation between presentation, public API, durable data, typed orchestration, background work, and narrow integrations.

Production topology

Layer	Production baseline	Responsibility
Web	Next.js on Vercel	Resident, manager, owner, tech, vendor, and operations surfaces; CDN and server components.
API	FastAPI on Cloud Run	Authentication, tenant context, idempotency, public API, SSE, and error handling.
Brain and workers	Cloud Run	Durable workflow, retrieval policy, evaluation, embeddings, rollups, reflection, and notifications.
Data	Supabase Postgres + pgvector, Redis, object storage	Tenant-scoped operational data, retrieval, queues/cache, signed media.
Integration	Narrow internal MCP services	Read data and execute only approved, scoped PMS, vendor, and notification actions.

The nine components

Component	Role	Write authority
Orchestrator	Durable state, routing, approval interruption, and execution.	Only component with side-effecting credentials.
Safety Sentinel	Life-safety determination.	None.
Intake Normalizer	Validated ticket facts from multimodal input.	None.
Context Broker	Evidence assembly and context budget.	None.
Diagnostician	Grounded cause and recommended action.	None.
Policy Auditor	Policy and statute/lease/SOP audit.	None.
Dispatch Planner	Vendor, parts, cost, and timing proposal.	None.
Communicator	Approved template rendering.	None.
Judge	Asynchronous run scoring.	None.

4. Engineering invariants

These are release blockers, not guidelines.

- No more than five model calls on the critical path. Independent reads run in parallel.
- Every table has organisation scope and row-level security. Every request sets tenant context.
- Decisions, model calls, guardrail events, and audit receipts are append-only.
- Protected attributes and health inferences are not stored, logged, or used as features.
- Every mutation requires an idempotency key. Long work returns a run ID and streams progress.
- Every material claim carries a citation ID or is put into an explicit unknowns channel.
- Durable facts supersede temporally; they are not overwritten. Durable memory promotion is controlled.
- Untrusted content is data, never instruction. Prompt injection is logged and surfaced as a security event.

Retrieval and memory

Knowledge is versioned Markdown with source ID, version, effective date, jurisdiction, authority, and tags. Retrieval combines lexical and vector search, rank fusion, metadata filters, measured optional reranking, parent expansion, and bounded extractive compression. The provenance envelope gives every model claim a source path.

Working state, authoritative entity facts, episodic retrieval, and versioned semantic knowledge are the four stores. Reflection is a controlled lifecycle job; a model cannot write durable memory on its own.

Security and failure

Failure	Safe response
P0 safety signal	Page the approved human path immediately; preserve the trigger record.
Schema or citation failure	One bounded repair attempt, then human review or abstention.
Retrieval miss	Broaden allowed filters once, declare policy not found, and escalate.
Provider or budget failure	Degrade through no-council, no-model, rules-only, then queue-and-acknowledge.
Tool timeout	Bounded retry and safe proposal-only fallback; never silently repeat side effects.

5. Evaluation and production operations

Evaluation is part of the product. Human labels are created before prompt tuning; the judge uses a pinned independent model family and version; CI uses tolerance bands rather than fragile exact thresholds.

Launch targets

Metric	Target	Gate
Ticket acknowledgement	99.9% under 1 second	Release SLO
P0 life-safety recall	At least 0.99	Hard regression gate
Priority macro-F1	At least 0.85	Tolerance-banded CI
Trade routing	At least 0.90	Golden-set gate
Groundedness	At least 0.95	Citation evaluation
Fair-housing leaks	Zero on red-team set	Hard release gate
Decision performance	p50 under 6s; p95 under 20s; cost under $0.05	Operations dashboard

The ML operations console

- A single time range aligns health, per-agent degradation, failures, model and prompt versions, calibration, cost, evaluation, and drift.
- Production failures can be promoted into a human label queue; nothing becomes golden data automatically.
- OpenTelemetry traces connect the request, evidence, model calls, guardrails, approval, and execution receipt.
- Alert on P0 regression, groundedness tolerance breach, escalation-rate jump, cost anomaly, failed execution receipt, and any fair-housing incident reaching a user.

Measured credibility

No target in this blueprint is a claim of current performance. Publish the evaluation dataset version, judge/prompt/model versions, run identifier, commit SHA, results, failures, and an explanation for every miss.

6. Delivery sequence

Later features never justify early scaffolding. Create a path only when the current phase uses it.

Phase	Goal	Exit condition
0. Contract freeze	Vocabulary, schemas, API/events, threat model, and test plan.	Contracts accepted together before feature code.
1. Foundation	Local tenant-isolated system with seed and API skeleton.	Migrations, RLS, auth, idempotency, and smoke tests pass.
2. Knowledge	Versioned corpus and retrieval before generation.	Retrieval evaluation and evidence view meet target.
3. Vertical slice	Safe resident-to-manager decision loop.	Seeded ticket runs from acknowledgement to trace.
4. Guardrails	Policy integrity and scoped execution.	Audit, guardrail, and fair-housing tests pass.
5. Evaluation	Measured MVP/recruiter checkpoint.	Public reproducible report and deliberately broken CI fixture.
6-8. Expand	Council, controlled memory, observability, notifications, community.	Measured lift and consent/safety controls.
9. Launch	Hardening, security, accessibility, operations.	Load/security/rollback and stranger-safe demo rehearsed.

Production release gate

- An unaided visitor can submit a synthetic ticket, see evidence and reasoning, approve it, and inspect the audit trace.
- Cross-tenant, P0, fair-housing, notification-suppression, provider-failure, and idempotency tests pass end to end.
- The evaluation report is reproducible and contains real numbers, including misses.
- Accessibility, security, dependency audit, secrets scan, load, observability, rollback, backup/restore, and incident procedures have been exercised.

Build the evidence-backed Maintenance Loop first. Everything else earns its place by making that loop safer, more measurable, or more useful.